

RAFFLES INSTITUTION

2013 Year 6 Preliminary Examination
Higher 1

CANDIDATE
NAME

--

CIVICS
GROUP

1	3	S	0	3	
---	---	---	---	---	--

INDEX
NUMBER

--	--	--	--	--	--

BIOLOGY

Paper 2 Core paper

8875/02

16th SEPTEMBER 2013

2 hours

Additional materials: Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **either ONE** question.

At the end of the examination, **hand in your essay SEPARATELY**.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/12
3	/8
4	/8
Section B	
5 OR 6	/20
Total	/60

This document consists of 10 printed pages.



Section A

Answer **all** the questions in this section.

- 1** **Fig. 1.1** shows the sequence of bases in a section of a gene coding for the first seven amino acids of a polypeptide chain.

AAACTGTTTCTGCTGGAACAT

Fig. 1.1

- (a) (i)** Methionine is coded by the initiation codon. On **Fig. 1.1**, label, the 3' and 5' end of the DNA template strand. [1]

- (ii)** List the corresponding codons in the mRNA transcribed from the template in **Fig. 1.1**.

..... [1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the **Table 1.1**.

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

Table 1.1

- (iii)** Using information in **Table 1.1**, work out the order of amino acids in the polypeptide.

..... [1]

- (b) (i)** Explain why single base deletions in the DNA sequence is more likely to produce non-functional proteins than single base substitutions.

.....

.....

.....

.....

.....

..... [3]

- (ii) Describe the effect of replacing guanine (underlined in **Fig.1.1**) with adenine.

.....

.....

.....

..... [2]

- (c) Explain how the structure of the nuclear envelope facilitates translation.

.....

.....

.....

..... [2]

- (d) A variety of related proteins can be produced from a gene. State the process and explain how the same gene can produce different protein products.

.....

.....

.....

..... [2]

[Total : 12]

- 2 An **absorption** spectrum is a graph of the absorption of different wavelengths of light by a photosynthetic pigment.
An **action** spectrum is a graph of the rate of photosynthesis at different wavelengths of light.
Fig. 2.1 shows the absorption spectra of chlorophyll a and chlorophyll b as well as an action spectrum.

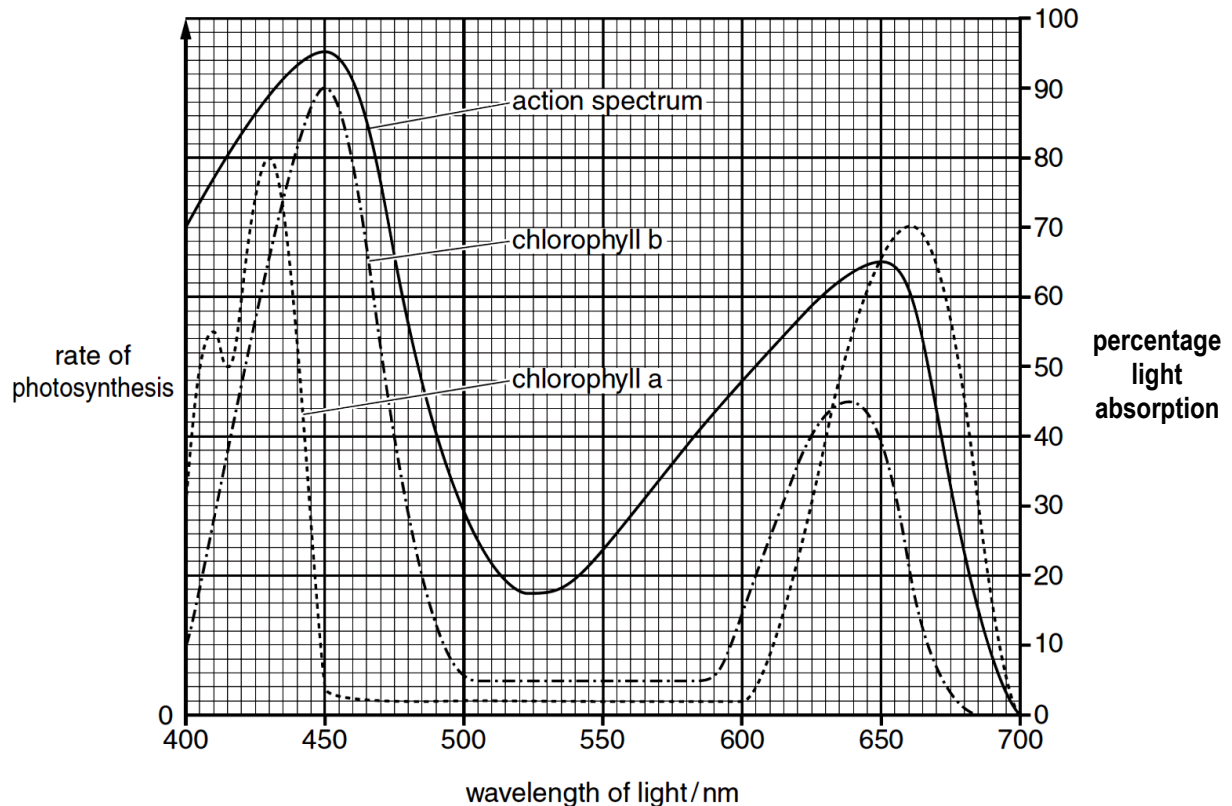


Fig. 2.1

- (a) With reference to **Fig. 2.1**,
(i) compare the **absorption** spectra of chlorophyll a and chlorophyll b.

.....

.....

.....

..... [2]

- (ii) explain the shape of the **action** spectrum.

.....

.....

.....

..... [2]

- (b) The light-dependent stage of photosynthesis takes place on the thylakoids of the chloroplast. **Fig. 2.2** represents a section through a thylakoid to show some of the components involved in the light-dependent stage.

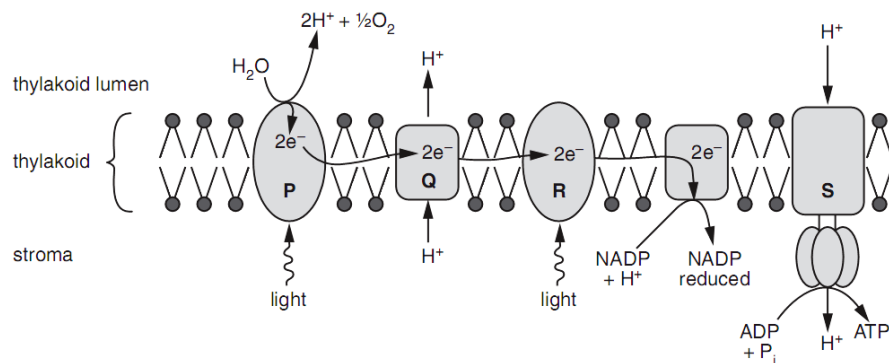


Fig. 2.2

- (i) Identify the structure labelled **S**.

S :

[1]

- (ii) Atrazine is a widely used herbicide (weed killer). It binds to a chloroplast protein involved in electron transfer between photosystems. Explain how atrazine works as an effective herbicide.

.....

.....

.....

.....

.....

..... [3]

- (c) In an investigation, mammalian liver cells were homogenised (broken up) and the resulting homogenate centrifuged. Samples of the complete homogenate and samples with only mitochondria were incubated with:

- 1 glucose
- 2 pyruvate
- 3 glucose and cyanide
- 4 pyruvate and cyanide

Cyanide inhibits oxidative phosphorylation.

After incubation the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in **Table 2.1**.

	Sample of homogenate			
	Complete		Only mitochondria	
	Carbon dioxide	Lactate	Carbon dioxide	Lactate
1 glucose	✓	✓	X	X
2 pyruvate	✓	✓	✓	X
3 glucose and cyanide	X	✓	X	X
4 pyruvate and cyanide	X	✓	X	X

X = absent ✓ = present

Table 2.1

- (i) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but **not** when they are incubated with glucose.

.....

.....

.....

..... [2]

- (ii) With reference to **Table 2.1**, explain the difference in the results when the complete cell homogenate is incubated with **2** and **4**.

.....

.....

.....

..... [2]

[Total : 12]

- 3** In cats, the gene controlling cat colour is sex-linked. This gene has two alleles.

In a particular colony of cats the coats of the males were either black or ginger and those of the females black, ginger or tortoiseshell (a patchwork black, ginger and cream).

- (a)** Using appropriate symbols, show the genotypes and their corresponding phenotypes of the different coats of the male cats mentioned above.

.....
.....[1]

- (b)** Using a genetic diagram, determine the phenotypes expected in the progeny of a cross between a black female and a ginger male.

[4]

- (c) In a litter of kittens, half the no. of females were tortoiseshell and the other $\frac{1}{2}$ were black; half the no. of males were black and the other half were ginger.

What were the colours of the parental male and female? Explain your answer.

.....

.....

.....

.....

.....

..... [3]

[Total : 8]

- 4 To create transgenic maize which can produce the *Bacillus thuringiensis* toxin, the recombinant plasmid shown in **Fig. 3.1** was constructed. The recombinant plasmid also contains the gene for phosphomannose isomerase (PMI), an enzyme which allows plants cells to utilize mannose as a carbon source.

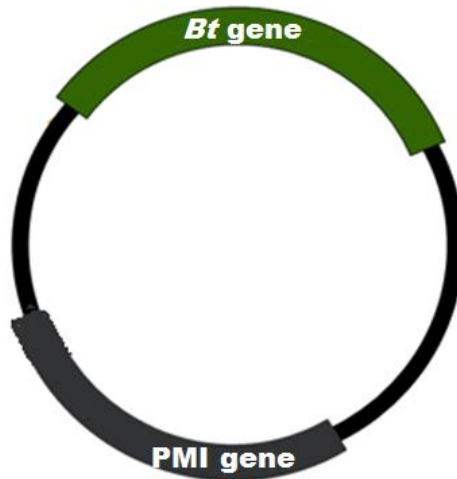


Fig. 4.1

- (a) Antibiotic or herbicide resistance genes are commonly included in the plasmid used for creating transgenic plants.

Explain why such genes were not needed in this case.

.....

.....

.....

..... [2]

To produce transgenic maize using the plasmid shown in **Fig. 4.1**, the procedure shown in **Fig. 4.2** was carried out.

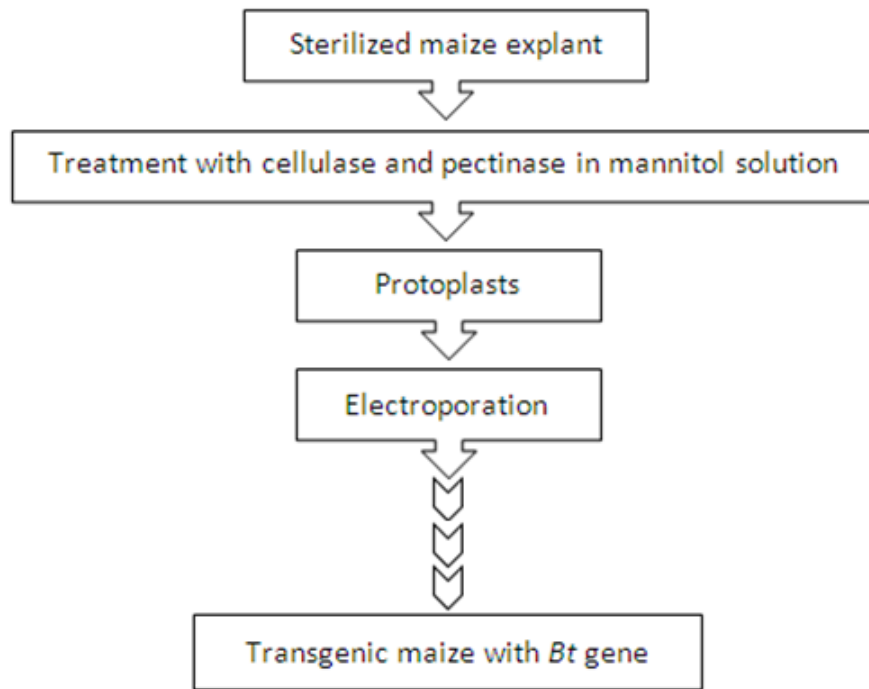


Fig. 4.2

With reference to **Fig. 4.2**,

- (b) (i) explain why the sterilized explant was treated with cellulase and pectinase before electroporation.

.....

 [2]

- (ii) Addition of mannitol to water lowers the water potential of water. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplasts are formed even if the sterilised explant was treated with cellulase and pectinase.

State what happens to the protoplasts if mannitol was not used.

.....
 [1]

- (c) Name an alternative method of introducing the *Bt* gene into the plant cells.

.....
 [1]

- (d) Explain the benefit of incorporating a *Bt* gene in the plasmid as shown in **Fig.4.1**.

.....

.....

..... [2]

[Total : 8]

Section B
Answer EITHER 5 OR 6.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Explain the effects of competitive and non-competitive inhibitors on the rate of enzyme activity. [8]
- (b)** Explain why variation is important in selection. [5]
- (c)** Explain how natural selection may bring about evolution. [7]

[Total: 20]

OR

- 6 (a)** Compare DNA polymerase and *Taq* polymerase. [5]
- (b)** Describe the properties of plasmids that allow them to be used as DNA cloning vectors. [6]
- (c)** Outline how insulin can be produced by genetic engineering. [9]

[Total: 20]

---- End of Paper ----